

By the end of this chapter, you'll be prepared to work with various types of datasets in different formats, and you'll have a deeper understanding of how to build complex visualizations. Being able to access and visualize online data is essential to working with a wide variety of real-world datasets.

The CSV File Format

One simple way to store data in a text file is to write the data as a series of values separated by commas, called *comma-separated values*. The resulting files are *CSV* files. For example, here's a chunk of weather data in CSV format:

```
"USW00025333","SITKA AIRPORT, AK US","2021-01-01",,"44","40"
```

This is an excerpt of weather data from January 1, 2021, in Sitka, Alaska. It includes the day's high and low temperatures, as well as a number of other measurements from that day. CSV files can be tedious for humans to read, but programs can process and extract information from them quickly and accurately.

We'll begin with a small set of CSV-formatted weather data recorded in Sitka; it is available in this book's resources at https://ehmatthes.github.io/pcc_3e. Make a folder called *weather_data* inside the folder where you're saving this chapter's programs. Copy the file *sitka_weather_07-2021_simple.csv* into this new folder. (After you download this book's resources, you'll have all the files you need for this project.)

NOTE

The weather data in this project was originally downloaded from <https://ncdc.noaa.gov/cdo-web>.

Parsing the CSV File Headers

Python's `csv` module in the standard library parses the lines in a CSV file and allows us to quickly extract the values we're interested in. Let's start by examining the first line of the file, which contains a series of headers for the data. These headers tell us what kind of information the data holds:

sitka_highs.py

```
from pathlib import Path
import csv

❶ path = Path('weather_data/sitka_weather_07-2021_simple.csv')
   lines = path.read_text().splitlines()

❷ reader = csv.reader(lines)
❸ header_row = next(reader)
   print(header_row)
```

We first import `Path` and the `csv` module. We then build a `Path` object that looks in the *weather_data* folder, and points to the specific weather data file we want to work with ❶. We read the file and chain the `splitlines()` method to get a list of all lines in the file, which we assign to `lines`.